

# Vertebra FEA Quickstart

Ogo spine compression workflow on the ogo-fea branch

This guide shows how to set up a new workstation and run a solved vertebra compression model with FAIM/N88. It assumes a density-calibrated CT image, a labelled vertebra segmentation, and access to the Numerics88 command-line tools.

## Workflow Summary

- Extract vertebral body and posterior process labels from the spine mask.
- Crop, ICP-align to the bundled Ogo vertebral reference, and resample to isotropic spacing.
- Build superior and inferior PMMA supports on stable body contact surfaces.
- Convert calibrated density to material IDs, write an .n88model, audit boundary conditions, solve with FAIM/N88, and write a compact results CSV.
- Optionally transform an ROI mask into model space and report masked Pistoia in addition to full-model Pistoia.

## Fresh Setup On A New Computer

### 1. Get Ogo

```
git clone git@github.com:Bonelab/Ogo.git
cd Ogo
git checkout ogo-fea
git pull origin ogo-fea
```

### 2. Create The Ogo Environment

```
conda env create -n ogo -f environment.yml
conda activate ogo
python -m pip install -e .
```

If dependency solving fails, create a plain environment and install from requirements.txt, then install Ogo editable:

```
conda create -n ogo python=3.10
conda activate ogo
python -m pip install -r requirements.txt
python -m pip install -e .

which ogoFEA
ogoFEA --help
```

## Make N88/FAIM Discoverable

Ogo builds the .n88model in Python, but solving and Pistoia require the Numerics88/FAIM command-line tools.

```
which n88modelinfo
which n88pistoia
which n88tabulate
which faim
```

| Setup                                            | How to run                                                                |
|--------------------------------------------------|---------------------------------------------------------------------------|
| N88 tools live in one directory                  | Pass --faim_bin_dir /path/to/n88/bin.                                     |
| N88 tools live in a separate conda environment   | Pass --faim_env ENV_NAME.                                                 |
| Individual tools have unusual names or locations | Pass explicit overrides such as --n88pistoia_command /path/to/n88pistoia. |

```
conda run -n ogoloco-n88 n88modelinfo --help
conda run -n ogoloco-n88 n88pistoia --help
```

## Required Inputs

| Input               | Requirement                                                                                 |
|---------------------|---------------------------------------------------------------------------------------------|
| Calibrated CT       | Scalar density image in K2HPO4-equivalent units.                                            |
| Labelled spine mask | Integer labels containing the vertebral body and posterior process.                         |
| Optional ROI mask   | Binary mask in the original CT image space, for example an L1 body mask for masked Pistoia. |

The vertebra target is passed as LEVEL:BODY\_LABEL:PROCESS\_LABEL. Example: L1:2:3 means output level L1, body label 2, and posterior process label 3.

## Verify With A Debug Model

```
ogoFEA spine \
  sub-001_desc-vqct_ct.nii.gz \
  sub-001_desc-spine_labels.nii.gz \
  --vertebra L1:2:3 \
  --output_path derivatives/fea_test \
  --debug \
  --no-solve
```

This checks image reading, labels, ICP alignment, PMMA support generation, and .n88model writing before launching a solve.

## Standard L1 Solve

```
ogoFEA spine \
  sub-001_desc-vqct_ct.nii.gz \
  sub-001_desc-spine_labels.nii.gz \
  --vertebra L1:2:3 \
  --output_path derivatives/fea \
  --threads 4 \
  --faim_bin_dir /path/to/n88/bin \
  --run_pistoia \
  --critical_volume 12 \
  --critical_strain 0.007
```

Use --threads 4 for a typical single-model run. This is the FAIM solver thread count for that one model; it does not launch multiple subjects in parallel.

## Full And Masked Pistoia

```
ogoFEA spine \
  sub-001_desc-vqct_ct.nii.gz \
  sub-001_desc-spine_labels.nii.gz \
  --vertebra L1:2:3 \
  --pistoia_mask sub-001_desc-L1Body_mask.nii.gz \
  --output_path derivatives/fea \
  --threads 4 \
  --faim_bin_dir /path/to/n88/bin \
  --run_pistoia \
  --critical_volume 12 \
  --critical_strain 0.007
```

| Result             | Description                                                                                 |
|--------------------|---------------------------------------------------------------------------------------------|
| Full-model Pistoia | Uses the solved vertebra model and excludes PMMA material.                                  |
| Masked Pistoia     | Uses only elements selected by the supplied ROI mask after transformation into model space. |

In one ogoFEA run, the same critical volume and critical strain are used for full-model and masked Pistoia. The results CSV includes masked\_pistoia\_\* fields when a mask is supplied.

## Expected Outputs

| File                     | Meaning                                                                                      |
|--------------------------|----------------------------------------------------------------------------------------------|
| *_L1.n88model            | Generated finite-element model.                                                              |
| *_L1_modeling.json       | Traceable metadata for inputs, preprocessing, materials, boundary conditions, and reporting. |
| *_L1_results.csv         | Compact one-row results table for downstream analysis.                                       |
| *_L1_pistoia.txt         | Full-model Pistoia report, when Pistoia is run.                                              |
| *_L1_masked_pistoia.txt  | ROI-masked Pistoia report, when --pistoia_mask is supplied.                                  |
| *_L1_pistoia_mask.nii.gz | ROI mask transformed into model space.                                                       |

## Model Defaults

| Setting                | Default                                                            |
|------------------------|--------------------------------------------------------------------|
| Model type             | Linear elastic vertebral compression.                              |
| High-level preset      | benchmark-linear.                                                  |
| Resampling             | 1.0 mm isotropic.                                                  |
| Density-to-modulus law | $E = 2980 * \rho^{1.05}$ MPa for benchmark-linear spine models.    |
| Bone Poisson ratio     | 0.3.                                                               |
| PMMA material          | $E = 2500$ MPa, Poisson ratio = 0.3.                               |
| PMMA geometry          | 10 mm thickness with stable-contact support placement.             |
| Default endpoint       | 0.68% compression strain, converted to millimeters before solving. |

## Quick Quality Checks

- The vertebral body should be upright in the model frame.
- The posterior process should be posterior to the body, not rotated into the axial direction.
- Superior and inferior PMMA supports should contact stable body surfaces rather than isolated osteophytes.
- The boundary-condition audit should pass.
- The modeling JSON should record the expected labels, material law, endpoint, and N88 settings.

## Common Problems

| Symptom                        | Likely cause                                        | What to check                                                       |
|--------------------------------|-----------------------------------------------------|---------------------------------------------------------------------|
| Cannot find faim or n88pistoia | N88 tools are not on PATH.                          | Pass --faim_bin_dir, --faim_env, or explicit command paths.         |
| Missing masked Pistoia fields  | No ROI mask was supplied, or masked Pistoia failed. | Check *_results.csv, *_masked_pistoia.txt, and the log.             |
| Process appears rotated        | ICP alignment or label definition problem.          | Run --debug --no-solve and inspect the model.                       |
| PMMA contacts osteophytes      | Endplate surface is irregular.                      | Inspect debug output and modeling JSON; exclude cases that fail QC. |
| Existing solution is reused    | The .n88model already contains a solution.          | Use a clean output directory before rerunning.                      |

## Minimal Reproducible Handoff

```
git clone git@github.com:Bonelab/Ogo.git
cd Ogo
git checkout ogo-fea
python -m pip install -e .
```

```
ogoFEA spine \
  image_K2HP04.nii.gz \
  spine_labels.nii.gz \
  --vertebra L1:2:3 \
  --pistoia_mask L1_body_mask.nii.gz \
  --output_path derivatives/fea \
  --threads 4 \
  --faim_bin_dir /path/to/n88/bin \
  --run_pistoia \
  --critical_volume 12 \
  --critical_strain 0.007
```

For tabulated analysis, use derivatives/fea/\*\_L1\_results.csv. For model inspection, use derivatives/fea/\*\_L1.n88model.